

# ABHI JAIN

Boston, MA

`abhijainstats.github.io` ♦ (+1)508-614-5822 ♦ `jaina22@bu.edu`

## EDUCATION

---

### **Boston University**

*September 2022 - May 2027 (expected)*

Ph.D. Biostatistics

- Dissertation: Bayesian tree-based methods for causal inference with spatial data
  - Advised by Dr. Shariq Mohammed and Dr. Sara Lodi
- Relevant coursework: Estimation Theory, Hypothesis Testing, Linear Models, Correlated Data Analysis, Nonparametric and Semiparametric Data Modeling, Statistical Analysis of Point Process Data, Causal Inference, Machine Learning

### **Wake Forest University**

*August 2020 - May 2022*

M.S. Statistics

- Master's Thesis: Using Zero-inflated Models to Estimate the Effect of Air Pollution on Self-Reported Mental Health
  - Advised by Dr. Rob Erhardt
- Relevant Coursework: Generalized Linear Models, Bayesian Statistics, Statistical Learning, Time Series, Mathematical Biology, Causal Inference, Stochastic Processes

### **Davidson College**

*August 2014 - May 2018*

B.A. Economics; Minor in Applied Mathematics, *cum laude*

- Relevant coursework: Microeconomic Theory, Macroeconomic Theory, Environmental Economics, Multivariable Calculus, Linear Algebra, Econometrics, Probability, Mathematical Statistics, Mathematical Modeling, Health Economics, Game Theory, Computational Economics

## EMPLOYMENT

---

### **Teaching Fellow**, Boston University School of Public Health

*May 2026 - Present*

- Support faculty in preparing education materials and integrating active learning and applied projects to Biostatistics courses and MS Biostatistics capstone
- Co-instructor for BS 800: Accelerated Statistical Training, which is a two week course for MS Applied Biostatistics students that covers probability, statistical inference, and computing in R and SAS
- Prepared slides and gave lectures on Hypothesis testing, Correlation and Regression, and Introduction to R for BU's Summer Institute of Biostatistics program as well as "Reproducibility in Research" and "Anatomy of a Journal Article" in Biostatistics Capstone course

### **Graduate Research Assistant**, Boston University School of Public Health *June 2022 - April 2026*

- Research in spatial statistics and modeling health outcomes
- Lead manuscript development; conduct literature reviews; perform data management and cleaning; statistical analysis tasks

### **Research Assistant**, Wake Forest Baptist Health

*February 2021 - May 2022*

- Conducted research into how healthcare usage has changed with Medicaid expansion by performing statistical analysis of large healthcare datasets in Stata
- Constructed datasets used to analyze hospitalization data; produced tables and figures to effectively communicate trends present in hospitalization data

#### Data Analyst, NORC

June 2018 - May 2022

- Served as a contractor for the Economics Unit of the Office of the Investor Advocate at the Securities and Exchange Commission.
- Developed econometric models to analyze investor decision-making and issues in financial markets
- Cleaned and performed statistical analysis on survey data and financial datasets

### RESEARCH INTERESTS

---

- **Methodology**  
Causal inference, causal machine learning, Bayesian statistics, Bayesian nonparametrics, spatial statistics, dynamic treatment regimes
- **Applications**  
Health policy, population health, spatial and environmental epidemiology, electronic health records

### PUBLICATIONS

---

1. **Jain A.**, LaValley M., Dukes K., Mohammed S. (2026) Modeling bounded well-being indices using Bayesian double generalized beta regression with spatial and temporal borrowing. *Spatial Statistics*.
2. **Jain A.**, Lugo N, Bautista-Hurtado D, Zuniga S, Penugonda A, Jean Philippe NJ, Labossiere S, Pluta KP, Mitchell S, Jack BW, Fernandez A, Drainoni ML, Salloum R, Austad K. (2026) A discrete choice experiment to understand hospital discharge preferences. *Accepted at Journal of General Internal Medicine*.
3. Alperen, J., LaValley, M., Ni, P., Tripp, T., Emig, D., Patil, P., Mohammed, S., **Jain A.**, Rickles M., Thompson E., Davis B., Sullivan S., Acker M., Dukes K. (2026) Characterizing U.S. Community Well-Being Indices: A Methodological Synthesis and Measurement Systems Perspective. *Population Health Metrics*.
4. Dukes, K., Ni, P., Alperen, J., Cesare, N., LaValley, M., Tripp, T., Lane, K., Mohammed, S., Patil, P., Emig, D., Winter, M., Davis B., Wang B., **Jain A.**, Acker M., and Rickles M. (2025) A national growth mixture modeling analysis of county-level COVID-19 incidence rate trajectories and health inequities during three successive pandemic waves in 2020. *Scientific Reports*.
5. **Jain A.**, LaValley M., Dukes K., Lane K., Winter M., Spangler K.R., Cesare N., Wang B., Rickles M., and Mohammed S. (2024) Modeling health and well-being measures using ZIP Code spatial neighborhood patterns. *Scientific Reports*.

### WORKING PAPERS

---

1. **Jain A.**, Jin W., Mohammed S., Lodi S., (2026+) Bayesian causal forests for joint treatments: an analysis of neighborhood food environments on obesity rates with interference. *In preparation*.
2. **Jain A.**, Mohammed S., Lodi S., Jin W., (2026+) Joint BART model with latent factors for causal inference in clustered observational studies. *In preparation*.
3. **Jain A.**, O'Donnell A., Madenci A., Lodi S., (2026+) Practical issues in generating a bootstrap distribution for causal effects of longitudinal treatment strategies. *In preparation*.

## CONFERENCE PRESENTATIONS

---

### Conferences (invited, contributed, poster)

1. Bayesian causal forests for estimating heterogeneous effects with joint treatments and interference. Contributed presentation at Joint Statistical Meetings, Boston, MA. *August 2026*
2. Modeling bounded well-being indices using Bayesian double-generalized beta regression with spatial and temporal borrowing. Contributed presentation at 2026 Eastern North American Region Spring Meeting, Indianapolis, USA. *March 2026*
3. Modeling well-being using ZIP Code spatial neighborhoods under a Bayesian Beta regression framework. Contributed presentation at 2025 Eastern North American Region Spring Meeting, New Orleans, USA. *2025 March*
4. Modeling well-being using ZIP Code spatial neighborhoods under a Bayesian Beta regression framework. Contributed presentation at Joint Statistical Meetings, Portland, USA. *August 2024*
5. Modeling health and well-being measures by incorporating zip-code spatial neighborhood patterns. Contributed presentation at Joint Statistical Meetings, Toronto, Canada. *August 2023*
6. Modeling health and well-being measures by incorporating zip-code spatial neighborhood patterns: a case study on Massachusetts and Georgia. Poster session presentation at New England Statistics Symposium, Boston, MA. *June 2023*
7. Regression-Based Oversampling for Mutual Fund Owners. Contributed poster presentation at Joint Statistics Meetings, Virtual. *August 2020*

### General audience

1. Bayesian causal forests for estimating heterogeneous effects with joint treatments and interference. Poster presentation at Boston University Population Health Data Science Poster Session, Boston, MA. *September 2026*
2. Performance of “bag of little bootstrap” in simulated complex longitudinal data. Kolokotronis Circle presentation at CAUSALab, Harvard T.H. Chan School of Public Health, Boston, MA. *September 2025*
3. Modeling well-being using ZIP Code spatial neighborhoods under a Bayesian Beta regression framework. Poster presentation at Boston University Population Health Data Science Poster Session, Boston, MA. *September 2025*
4. Modeling health and well-being measures by incorporating zip-code spatial neighborhood patterns. Poster presentation at Boston University Population Health Data Science Poster Session, Boston, MA. *September 2023*

## TEACHING EXPERIENCE

---

### Teaching Fellow

Department of Biostatistics, Boston University School of Public Health

- SPH BS 880 — Biostatistics Capstone: Design and Analysis of Investigations *Fall 2026*
- SPH BS 755 — Theory of Linear Models in Biostatistics *Fall 2026*
- SPH BS 800 — Accelerated Statistical Training *Summer 2026*
- SPH BS 401S — Survey in Biostatistical Methods *Summer 2026*

## Teaching Assistant

Department of Biostatistics, Boston University School of Public Health

- SPH BS 849 — Bayesian Modeling for Biomedical Research & Public Health *Spring 2026*
- SPH BS 807 — Applied Causal Inference in Health Research *Fall 2024*

Department of Mathematics and Statistics, Wake Forest University

- STA 111 — Elementary Probability and Statistics *Fall 2020*

Department of Economics, Davidson College

- ECO 105 — Statistics and Basic Econometrics *Spring 2018*

## Tutor

Department of Mathematics and Statistics, Wake Forest University

- Math & Stats Center *Fall 2020*

## HONORS & AWARDS

---

- Distinguished Student Paper Award - *ENAR 2026 Spring Meeting*
- Student Travel Award (Fall 2024) - *Boston University School of Public Health*

## SERVICE

---

|                                                                                |                                    |
|--------------------------------------------------------------------------------|------------------------------------|
| <b>Session Chair</b> - Bayesian Methods for Categorical and Survey Data at JSM | <i>August 2026</i>                 |
| <b>Officer</b> - Boston University Student Chapter of the ASA                  | <i>March 2024-Present</i>          |
| <b>Student representative</b> - Biostatistics Graduate Education Committee     | <i>June 2024-Present</i>           |
| <b>Volunteer academic mentor</b> - Boston Partners in Education                | <i>February 2024-December 2025</i> |
| <b>Lead student volunteer</b> - New England Statistics Symposium               | <i>May 2023</i>                    |

## SKILLS

---

### Programming Languages and Software

R, Stata, Python, SAS, SQL, GitHub, Excel, Word, PowerPoint, Visio, Final Cut Pro

### Languages

Hindi (speaking)